

KOMMU SAI PRANEETH

Data Engineer

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PROFESSIONAL SUMMARY

- 6+ Years of Experienced in designing and developing robust **data pipelines** using **Python, PySpark**, and **SQL** to support large-scale data processing across cloud and on-premise environments for various industry domains.
- Proficient in working with **big data technologies** such as **Apache Spark, Kafka**, and the **Hadoop Ecosystem (HDFS, Hive, MapReduce)** for processing, analyzing, and transforming large volumes of data efficiently.
- Expert in implementation of Azure Data solutions, **ETL**, data migration solutions using **Azure Databricks**, Azure Data Factory, Azure Data Lake, and SQL databases
- Skilled in deploying and managing cloud-based data platforms on **AWS, Azure**, and **Google Cloud Platform**, leveraging services like **AWS Glue, Azure Synapse**, and **BigQuery** for scalable storage and compute.
- Adept in programming languages like **Python** including **Big Data technologies** like Hadoop, Hive.
- Lead projects involving Installing, Configuring, Patching, Upgrading **SQL Server 2014, 2017, 2019, 2022** and **data migration** among versions.
- Strong expertise in **containerization** and **orchestration** using **Docker** and **Kubernetes** for packaging and deploying microservices-based data processing applications in distributed environments.
- Experienced in designing and implementing secure **RESTful APIs** using **OAuth2/JWT** standards, enabling safe data exchange between internal systems and third-party applications.
- Deep understanding of data warehousing principles and expertise building optimized data warehouses using **Amazon Redshift, Azure Synapse** and **BigQuery** for business intelligence and reporting.
- Skilled in developing real-time streaming data solutions with **Apache Kafka, Spark Streaming**, and **Google Pub/Sub** to deliver low-latency data analytics and operational insights.
- Expertise in building complex **Extract, Transform and Load** ETL packages using varied and complex data sources and merging them into a single data model.
- Knowledgeable in implementing **data governance** and **data quality** frameworks such as **Great Expectations** and **Apache Ranger** to ensure compliance and accuracy across data pipelines.
- Experienced in setting up **CI/CD pipelines** for data projects using **Jenkins, Azure DevOps**, and **Git**, enabling seamless continuous integration and deployment processes.
- Proficient in monitoring and alerting using tools like **Prometheus, Grafana, CloudWatch**, and **Azure Monitor** to ensure pipeline health and performance in production environments.
- Extensively worked with **Avro** and **Parquet** files and converted the data from either format Parsed Semi Structured **JSON** data and converted to Parquet using Data Frames in **PySpark**.
- Passionate about mentoring junior engineers, conducting detailed code reviews, and promoting best practices in software development and data engineering.
- Expertise in **OLTP** and **OLAP** data modeling, SQL performance tuning techniques and normalization process.
- Collaborative team player, adept at working closely with data scientists, business analysts, and stakeholders to translate complex business requirements into scalable data engineering solutions.
- Strong analytical skills to optimize **SQL queries** and data processing workflows for better performance, cost savings, and faster report generation.
- Hands-on experience across the complete **Software Development Life Cycle (SDLC)** using **Agile** and hybrid methodologies, with active contributions to sprint planning, retrospectives, and backlog grooming.
- Excellent communication skills, capable of presenting technical concepts to non-technical stakeholders.

TECHNICAL SKILLS

Category	Skills
Programming Languages	Python, SQL, PySpark, Scala, T-SQL, R, Unix/Linux, Shell Scripting
Big Data Tools	Apache Spark, Apache Kafka, Hadoop Ecosystem (HDFS, MapReduce, Hive, HBase), Spark Streaming, Apache NiFi
Cloud Technologies	Azure, Azure Data Factory, Azure Databricks, Azure Synapse Analytics, Azure HDInsight AWS, S3, Redshift, AWS Glue, Google Cloud, BigQuery
Data Warehousing & Databases	Snowflake, Data Lakes, Azure Data Lake Storage, AWS Redshift, Google BigQuery Azure SQL Database, Azure Cosmos DB, MS SQL Server, PostgreSQL, MySQL, MongoDB
Data Visualization & Analysis	Microsoft Excel, Microsoft Power BI, Tableau, Looker, AWS QuickSight

Data Modeling	Star & Snowflake Schema, OLTP, OLAP, Data Management
ETL Tools & Data Integration	Apache Airflow, Informatica, AWS Glue, Azure Events Hubs, Apache Kafka, Data Pipelines
Machine Learning & MLOps	Azure ML, Scikit-learn, TensorFlow, SageMaker, XGBoost, Gen AI (OpenAI)
API & Microservices	REST API, JWT/OAuth2 Authentication, API Gateways, Microservices Architecture
Data Administration	Installation & Configuration, Performance Tuning, Backup & Recovery, High Availability, Log Shipping, Mirroring, Replication, Clustering, Activity Monitor, Event Viewer
Monitoring & DevOps	Prometheus, Grafana, Azure Monitor, CloudWatch, CI/CD, Docker, Kubernetes (AKS), Jenkins, Git, Github, Agile Methodologies
Soft Skills	Communication, Cross-Functional Collaboration, Analytical Thinking, Business Acumen

PROFESSIONAL EXPERIENCE

Client: Bank of America

Senior Data Engineer | May 2024 – Present

- Designed and implemented highly scalable **data ingestion pipelines** utilizing **Apache Kafka**, **Python**, and **Spark Streaming** to process millions of financial transactions daily, ensuring real-time data availability and low-latency processing for fraud detection.
- Developed complex batch and streaming **ETL workflows** with **PySpark** and orchestrated them using **Apache Airflow** for automated scheduling, error handling, and alerting, improving pipeline reliability and reducing downtime.
- Implemented comprehensive **data governance** policies with **Apache Ranger** and integrated **Kerberos authentication** to secure sensitive customer data, ensuring compliance with regulations such as GDPR and CCPA in banking systems.
- Built containerized microservices for data processing using **Docker** and deployed them on **Kubernetes clusters**, enabling flexible and scalable data transformation services aligned with cloud-native architecture principles.
- Integrated machine learning models into production pipelines by leveraging **MLflow** for model tracking and deployment, enabling predictive analytics to enhance credit risk assessment and customer segmentation processes.
- Developed and maintained secure, high-performance **RESTful APIs** with **OAuth2** authentication to expose processed data to internal applications and authorized external clients for real-time analytics and reporting.
- Delivered optimized **SQL/PySpark** outputs integrated with **Power BI dashboards** for financial reporting.
- Automated continuous integration and delivery of data pipelines using **Jenkins** and managed source control with **Git**, facilitating faster deployments and reducing manual errors in the production environment.
- Created comprehensive monitoring dashboards using **Grafana** integrated with **Prometheus** metrics to track pipeline health and data latency, providing timely alerts and enabling proactive troubleshooting.
- Collaborated closely with business analysts and data scientists to translate complex financial reporting requirements into optimized **SQL** queries and **PySpark** scripts for analytical dashboards.
- Led the migration of legacy **Hadoop** jobs to cloud-native data platforms, utilizing **Azure Data Factory**, **Azure Databricks** and **HDInsight** to improve scalability, reduce operational overhead, and streamline maintenance efforts.
- Implemented automated data validation and quality checks using **Great Expectations**, ensuring data integrity and consistency across multi-stage data pipelines that support critical banking applications.
- Worked with security teams to conduct regular vulnerability assessments on data infrastructure, implementing encryption at rest and in transit using **TLS** and role-based access controls to protect sensitive information.
- Enhanced the data lake architecture by leveraging **Azure Data Lake Storage Gen2** storage with lifecycle policies and data tiering, reducing storage costs and improving data accessibility for analytics teams.
- Delivered knowledge transfer sessions on cloud-native data engineering practices, container orchestration, and pipeline optimization, improving team skill sets and accelerating project delivery timelines.
- Mentored junior engineers in developing efficient **Spark** jobs, debugging complex data processing issues, and adopting DevOps best practices, resulting in improved overall team productivity and quality of deliverables.

Client: Cholamandalam MS General Insurance

Data Engineer | Aug 2021 – Aug 2023

- Designed and maintained scalable **ETL pipelines** using **Python**, **Apache Spark**, and **SQL** to efficiently process high-volume insurance claims data, enabling accurate and timely reporting for operational decision-making.
- Developed real-time streaming solutions by integrating **Apache Kafka** with **Spark Streaming** to monitor claims processing workflows, significantly reducing data latency and improving fraud detection capabilities.
- Architected and optimized cloud-based **data warehousing** solutions using **Azure Synapse Analytics**, enabling fast and cost-effective querying of large datasets for business intelligence and analytics teams.
- Automated infrastructure provisioning and data pipeline deployments using **Terraform** and **Azure DevOps CI/CD**, resulting in consistent, repeatable deployments and reducing manual configuration errors.
- Implemented strict data security and compliance controls using **Azure IAM** and **Azure Purview APIs** to protect sensitive insurance data and maintain regulatory compliance.

- Containerized data ingestion and transformation microservices using **Docker**, orchestrating deployments with **Kubernetes** clusters to ensure scalability, high availability, and ease of management.
- Building **KPI's**, measures, calculated columns and calculated tables using **Data analysis expressions (DAX)**. That helped users to use the **PowerBI** in a Self-Service way.
- Integrated machine learning models into production pipelines through **Kubeflow**, enabling predictive analytics for identifying fraudulent claims and optimizing underwriting processes.
- Designed and maintained **Azure Data Factory** and **Apache Airflow** workflows for orchestrating complex **ETL** jobs, improving pipeline transparency, monitoring, and error recovery.
- Optimized **SQL queries** and database indexing to enhance performance of reporting applications and reduce resource consumption on cloud-based data warehouses.
- Collaborated with data scientists and analysts to prepare and clean large datasets using **PySpark** and **Pandas**, facilitating accurate and efficient model training.
- Established automated data quality checks using **Great Expectations**, ensuring that incoming data met business-defined standards for accuracy and completeness.
- Developed secure and scalable REST APIs adhering to **RESTful principles** and implemented **JWT-based authentication** for data access control.
- Automated routine system and data management tasks using **Unix shell scripting**, enhancing operational efficiency and reducing human errors.
- Engaged regularly with cross-functional stakeholders to gather evolving requirements and deliver technical solutions aligned with business goals.
- Documented data engineering processes, architecture designs, and workflows thoroughly, supporting knowledge sharing and onboarding of new team members.
- Led training sessions on cloud data engineering tools, container orchestration, and data pipeline best practices, improving team competencies and delivery speed.

Client: Trinity Health

Data Engineer | Jun 2018 – Aug 2021

- Developed end-to-end **data pipelines** using **Python**, **SQL**, and **Apache Spark**, efficiently processing electronic health records (EHR) and claims data to support clinical analytics and reporting initiatives.
- Built and optimized batch processing workflows in **Apache Airflow**, enabling automated scheduling, monitoring, and failure recovery for critical healthcare data jobs.
- Leveraged **Azure Data Factory** to orchestrate data ingestion from multiple on-premises and cloud sources, ensuring reliable and timely availability of healthcare data.
- Created scalable **data warehouse** solutions on **Microsoft Azure Synapse**, integrating diverse healthcare datasets for unified analytics and improved patient care insights.
- Implemented robust data quality frameworks using **Great Expectations**, ensuring integrity and accuracy of clinical and operational data consumed by analysts and clinicians.
- Optimized SQL queries and database schemas to improve performance of complex healthcare analytics reports, reducing latency and enhancing user experience.
- Utilized **Power BI** in collaboration with BI teams to create interactive dashboards visualizing patient outcomes, operational metrics, and financial performance.
- Developed secure **REST APIs** for exposing aggregated healthcare data to internal applications with **OAuth2** based authentication, enhancing interoperability across systems.
- Configured **Azure Monitor** and **Log Analytics** to track data pipeline health, performance, and errors, enabling proactive issue resolution and minimizing downtime.
- Collaborated with clinical data scientists to prepare and clean datasets for advanced machine learning models predicting patient readmissions and treatment efficacy.
- Involved in knowledge sharing sessions on cloud data engineering tools and best practices, enabling the team to adopt scalable and maintainable solutions.
- Participated in Agile ceremonies, contributing to sprint planning, backlog grooming, and retrospectives to ensure timely delivery of data engineering tasks.
- Assisted in the migration of legacy on-premise databases to cloud platforms, improving scalability, data accessibility, and disaster recovery capabilities.

EDUCATION

Master of Science (M.S.) in Computer Science

New Jersey Institute of Technology, *Newark, New Jersey*

CERTIFICATION

- Microsoft Certified: DP 900 - Azure Data Fundamentals